

ORIGINAL ARTICLE

GABRP regulates chemokine signalling, macrophage recruitment and tumour progression in pancreatic cancer through tuning KCNN4-mediated Ca^{2+} signalling in a GABA-independent manner

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► Additional material is published online only. To view, please visit the journal online (<http://dx.doi.org/10.1136/gutjnl-2018-317479>).

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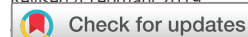
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Received 27 August 2018

Revised 4 February 2019



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To cite: Jiang S-H, Zhu L-L, Zhang M, et al. *Gut* Epub ahead of print: [please include Day Month Year]. doi:10.1136/gutjnl-2018-317479

ABSTRACT

Background and aims Pancreatic ductal adenocarcinoma (PDAC) is a leading cause of cancer-related death worldwide. Neurotransmitter-initiated signalling pathway is profoundly implicated in tumour initiation and progression. Here, we investigated whether dysregulated neurotransmitter receptors play a role during pancreatic tumorigenesis.

Methods The Cancer Genome Atlas and Gene Expression Omnibus datasets were used to identify differentially expressed neurotransmitter receptors. The expression pattern of gamma-aminobutyric acid type A receptor pi subunit (GABRP) in human and mouse PDAC tissues and cells was studied by immunohistochemistry and western blot analysis. The in vivo implications of GABRP in PDAC were tested by subcutaneous xenograft model and lung metastasis model. Bioinformatics analysis, transwell experiment and orthotopic xenograft model were used to identify the in vitro and in vivo effects of GABRP on macrophages in PDAC. ELISA, co-immunoprecipitation, proximity ligation assay, electrophysiology, promoter luciferase activity and quantitative real-time PCR analyses were used to identify molecular mechanism.

Results GABRP expression was remarkably increased in PDAC tissues and associated with poor prognosis, contributed to tumour growth and metastasis. GABRP was correlated with macrophage infiltration in PDAC and pharmacological deletion of macrophages largely abrogated the oncogenic functions of GABRP in PDAC. Mechanistically, GABRP interacted with KCNN4 to induce Ca^{2+} entry, which leads to activation of nuclear factor κB signalling and ultimately facilitates macrophage infiltration by inducing CXCL5 and CCL20 expression.

Conclusions Overexpressed GABRP exhibits an immunomodulatory role in PDAC in a neurotransmitter-independent manner. Targeting GABRP or its interaction partner KCNN4 may be an effective therapeutic strategy for PDAC.

Significance of this study

What is already known on this subject?

- The tumour microenvironment of pancreatic ductal adenocarcinoma (PDAC) is a rich resource for neurotrophic factors and neurotransmitters.
- Neurotransmitter-initiated signalling pathway is frequently activated in human cancers, including PDAC.
- Dysregulated gamma-aminobutyric acid type A receptor pi subunit (GABRP) expression is associated with malignant phenotypes in several cancers.

What are the new findings?

- GABRP is increased during malignant transformation and predicts a poor prognosis in PDAC.
- GABRP contributes to PDAC tumour growth in macrophage-dependent and gamma-aminobutyric acid-independent manner.
- GABRP promotes macrophage recruitment by inducing CXCL5 and CCL20 expression.
- GABRP modulates CXCL5 and CCL20 expression through KCNN4/ Ca^{2+} /nuclear factor κB axis.

How might it impact on clinical practice in the foreseeable future?

- As GABRP is increased in PDAC and predicts poor prognosis, strategies for targeting GABRP may hold distinct promise as prospective clinical therapies.

INTRODUCTION

Similar to the processes of neoangiogenesis and lymphangiogenesis, the possibility of formation of new nerve endings within the tumours, a phenomenon known as neoneurogenesis was widely identified.^{1–4} This process further expanded our

knowledge of the complex network of neurotransmitters related to tumour progression. Indeed, neurotransmitters and neuropeptides can promote tumour progression by stimulating the migration and invasion ability of cancer cells, a key event in the dissemination of tumour cells to distant sites.^{5,6} Pancreatic ductal adenocarcinoma (PDAC) as one of the most aggressive human malignancies is characterised by pronounced tumour desmoplasia, reduction in intratumoural vessel density and neural remodelling.^{7,8} The tumour microenvironment of PDAC is a rich resource for neurotrophic factors such as glial cell derived neurotrophic factor (GDNF), nerve growth factor (NGF), artemin and autonomic neurotransmitter (epinephrine and norepinephrine), all of which can facilitate the growth or invasiveness of PDAC cells.^{9,10} In this study, we aimed to investigate the dysregulated neurotransmitter receptors and neuroactive ligand receptors expressed by PDAC cells and uncover its/their oncogenic functions in PDAC.

Gamma-aminobutyric acid (GABA) is the major inhibitory neurotransmitter in the central nervous system, but it is also found to widely distribute in the peripheral endocrine organs such as pituitary, adrenal medulla and GI tract.¹¹ Three cellular receptors of GABA have been identified: the ionotropic GABA_A and GABA_C receptors and the metabotropic GABA_B receptor.^{11,12} The oligomeric chloride channel GABA_A and GABA_C are heteromeric complexes composed of five subunits. GABRP is the π subunit of the GABA_A receptor and controls the sensitivity of recombinant receptors to modulatory agents.¹³ In this study, we identified the differentially expressed GABRP as a tumour promoter in PDAC. Previously, several reports have documented the oncogenic activities of GABRP in cancers, including breast cancer, pancreatic cancer and ovarian cancer.^{14,15} However, the underlying mechanisms of GABRP in cancers remain largely unknown. Here, we further sought to delineate the mechanisms of GABRP in PDAC tumorigenesis.

MATERIALS AND METHODS

Tissue specimens

A tissue microarray containing 311 pathologist-certified and clinically annotated PDAC specimens and corresponding non-cancerous tissues were obtained from Ren Ji Hospital from January 2002 to June 2015. All the patients enrolled in this study provided written informed consent.

Animal models

The widely available genetically engineered mouse model, *Kras*^{G12D/+}; *Trp53*^{R172H/+}; *Pdx1-Cre*; (KPC), was used in this study to acquire the precursor lesions (pancreatic intraepithelial neoplasia [PanINs]) of PDAC in mouse. The *Pdx1-Cre* and *LSL-Trp53*^{R172H/+} mice were all obtained from The Jackson Laboratory (<https://www.jax.org>). The *LSL-Kras*^{G12D/+} mice were a generous gift from Professor Xiu-Feng Pang (Shanghai Key Laboratory of Regulatory Biology, Institute of Biomedical Sciences and School of Life Sciences, East China Normal University). All animal experiments were undertaken in accordance with the NIH Guide for the Care and Use of Laboratory Animals.

Additional protocols and procedures are described in online supplementary materials and methods.

Statistical analysis

Data were presented as the means $\pm$ SD. Statistical analyses were done using SPSS V.16.0 for Windows (IBM) or GraphPad Prism (GraphPad Software, San Diego, California, USA). Cumulative survival time was calculated by the Kaplan-Meier method and

analysed by the log-rank test. Correlation of GABRP expression with clinical parameters in patients with PDAC was evaluated by χ^2 test. Correlation between GABRP and neighbour genes was determined using the Spearman's test. The student's t-test or one-way ANOVA was used for comparison between groups. P values <0.05 were considered to be statistically significant.

RESULTS

GABRP is increased during malignant transformation and predicts a poor prognosis in PDAC

To identify the potential neurotransmitter receptors dysregulated in PDAC, we first analysed the Gene Expression Omnibus (GEO) dataset (GSE15471)¹⁶ containing expression profile data of 36 microdissected PDAC and matching normal pancreatic tissue samples. Based on Kyoto Encyclopedia of Genes and Genomes (KEGG) database, 106 neurotransmitter receptors of 8 classical neurotransmitter systems and 146 neuroactive ligand receptors were submitted to analysis (online supplementary table 1). Of these, three receptors had a fold change larger than 4, named gamma-aminobutyric acid type A receptor π subunit (GABRP), endothelin receptor type A (EDNRA) and coagulation factor II thrombin receptor (F2R) (figure 1A). The expression profile of these three receptors in PDAC was further confirmed in another two PDAC cohorts (GSE16515 and GSE28735),^{17,18} which also contain paired tumour and non-tumour specimens (figure 1B). PanINs is a classical precursor lesion of PDAC.¹⁹ Interestingly, we found that GABRP expression was remarkably increased in human PanINs compared with normal pancreas tissues (figure 1C), whereas EDNRA and F2R expression remained largely unaltered. To test the prognostic value of these aberrant receptors, data derived from The Cancer Genome Atlas (TCGA) cohort with clinical follow-up information were used for Kaplan-Meier analysis. Of note, patients with PDAC with higher GABRP expression had a poor prognosis compared with those with a lower GABRP expression (figure 1D). No significant difference in the survival time was found based on EDNRA or F2R expression (online supplementary figure 1A) and we therefore focused on the study of GABRP in PDAC.

To further determine the expression pattern of GABRP protein in PDAC, we performed immunohistochemical analysis in a preclinical PDAC mouse model (KPC) and a large-scale human PDAC tissue microarray. As a result, GABRP immunoreactivity was increased from normal mouse pancreas to PanINs and PDAC in a stepwise manner (figure 1E and online supplementary figure 1B). In human, GABRP was strictly expressed in the islets of normal pancreas, but commonly detected in the PDAC cells (figure 1F and online supplementary figure 1C). To explore the prognostic value of GABRP proteins in our cohort, we analysed overall survival (OS) of patients with PDAC with GABRP expression. The result showed that patients with GABRP high-expression have significantly shorter OS time than those with GABRP low-expression (figure 1F).

GABRP promotes tumour growth and metastasis in PDAC

To elucidate the biological functions of GABRP in PDAC, we first compared the gene expression profiles between GABRP-low group and GABRP-high group by using the RNA sequencing (RNA-seq) data of 179 patients with PDAC in the TCGA cohort. Gene set enrichment analysis revealed that the gene sets related to cell proliferation and metastasis, including FIRESTEIN_Proliferation, JAEGER_Metastasis_Up and NAKAMURA_Metastasis, were enriched in samples with high GABRP expression (figure 2A). To gain

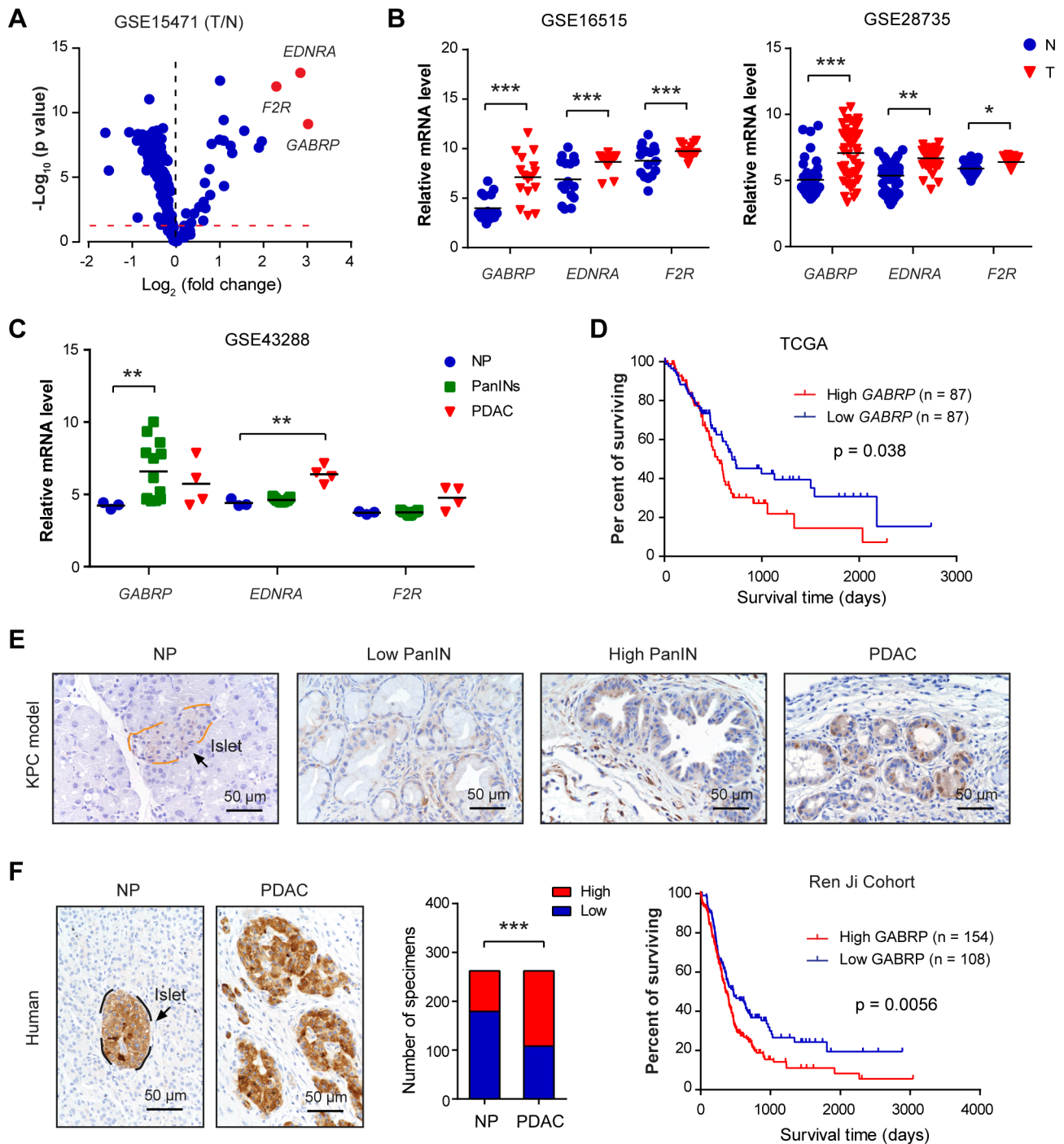


Figure 1 Gamma-aminobutyric acid type A receptor pi subunit (GABRP) is increased during malignant transformation and predicts a poor prognosis in pancreatic ductal adenocarcinoma (PDAC). (A) Volcano plot showing fold changes (x axis) and corresponding p values ($\log_{10}$, y axis) of normal vs pancreatic cancer samples. Data were obtained from GSE15471. Red dot represents fold change values >4 . (B) Expression analysis of *GABRP*, endothelin receptor type A (*EDNRA*) and coagulation factor II thrombin receptor (*F2R*) in tumours (T) and matching normal (N) pancreatic tissue samples using two independent cohorts (GSE16515, $n=16$; GSE28735, $n=45$) derived from Gene Expression Omnibus datasets. (C) Expression analysis of *GABRP*, *EDNRA* and *F2R* in the precursor lesions of pancreatic cancer from GSE43288 dataset. NP, normal pancreas; PanINs, pancreatic intraepithelial neoplasia. (D) Kaplan-Meier analysis (log-rank test) of the overall survival of patients with PDAC based on *GABRP* expression. Data were derived from The Cancer Genome Atlas (TCGA) cohort. (E) Representative immunohistochemical images showing GABRP expression from normal pancreas to PanINs and PDAC of *Kras*^{G12D/+}; *Trp53*^{R172H/+}; *Pdx1-Cre* (KPC) mice. Scale bar: 50 μ m. (F) Immunohistochemical analysis of GABRP expression in human PDAC tissue microarray. Representative GABRP images were shown at left panel. The percentage of tissue cores displaying low and high staining in normal pancreas and PDAC lesions was shown at middle panel. The correlation between GABRP expression and patients' overall survival was conducted in Ren Ji cohort (right panel). Scale bar: 50 μ m. * $P<0.05$, ** $p<0.01$, *** $p<0.001$.

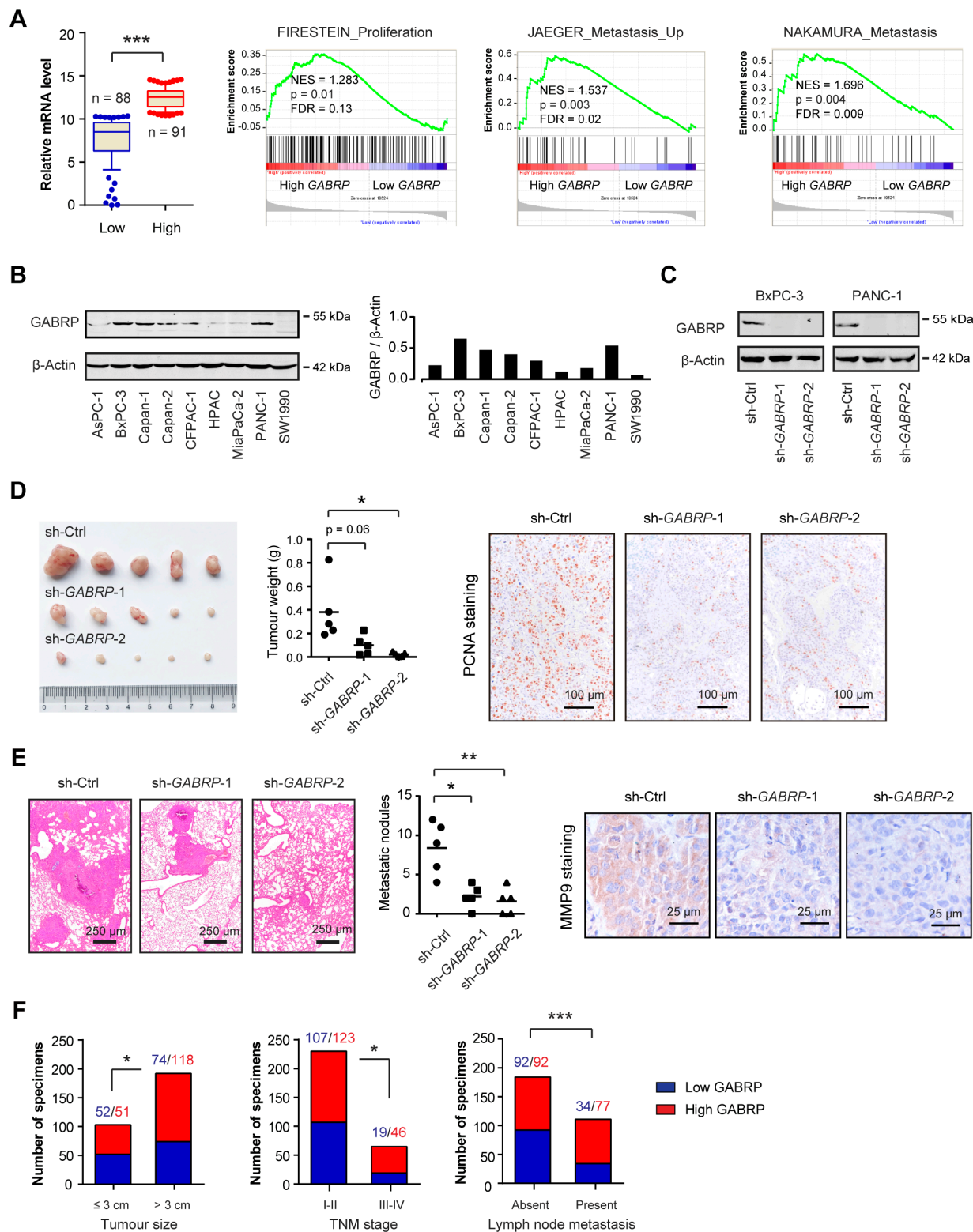


Figure 2 Gamma-aminobutyric acid type A receptor pi subunit (GABRP) promotes tumour growth and metastasis in pancreatic ductal adenocarcinoma (PDAC). (A) The Cancer Genome Atlas (TCGA) RNA sequencing data were divided into two groups based on *GABRP* expression. Gene set enrichment analysis plot based on the gene expression profiles of *GABRP*-high group compared with *GABRP*-low group. NES, normalised enrichment score. False discovery rate (FDR) was set at 0.25. (B) Western blot analysis of GABRP protein expression in PDAC cells. (C) Western blot analysis of short hairpin (sh) RNA-mediated GABRP knockdown efficiency in BxPC-3 and PANC-1 cells. (D) BALB/c nude mice were subcutaneously implanted with 2×10^6 sh-Ctrl and sh-GABRP BxPC-3 cells (n=5). Four weeks later, tumour burden was measured. Representative immunohistochemical images of PCNA staining from sh-Ctrl and sh-GABRP subcutaneous xenograft tissues were shown in the right panel. Scale bar: 100 μ m. (E) The sh-Ctrl and sh-GABRP BxPC-3 cells were injected into the tail vein of BALB/c nude mice. After 30 days, the lung metastasis and matrix metalloproteinase 9 (MMP9) expression were determined. (F) Correlation analysis of GABRP expression with tumour size, tumour, node, metastases (TNM) stages and lymph node metastasis of patients with PDAC from Ren Ji cohort (Fisher's exact test). * $P < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

further insight into the oncogenic roles of GABRP in PDAC, we performed short hairpin RNA (shRNA)-mediated loss-of-function study in PDAC cells. Two cell lines with high GABRP protein levels, BxPC-3 and PANC-1, were selected for functional analysis (figure 2B and online supplementary figure 1D). Stable expression of two shRNA (sh-GABRP-1, sh-GABRP-2) targeting *GABRP* resulted in >80% decrease in GABRP protein expression (figure 2C). In subcutaneous xenograft model, the tumour growth of PDAC in sh-GABRP group (sh-1 and sh-2) was significantly reduced compared with the sh-Ctrl group as revealed by tumour weight and the proliferation index proliferating cell nuclear antigen (PCNA) staining (figure 2D). Moreover, we noticed that knockdown of GABRP also suppressed *in vivo* lung metastasis of PDAC cells (figure 2E). Consistently, GABRP expression was closely associated with the clinicopathological parameters in our cohort, including tumour size, tumour, node, metastases stage and lymph node metastasis (figure 2F). Therefore, these data suggest that GABRP is profoundly implicated in the malignant phenotypes of pancreatic cancer.

GABRP correlates with macrophage infiltration in PDAC

To interrogate the molecular and cellular basis by which GABRP promotes tumour progression, we annotated the cellular functions of 300 *GABRP*-correlated genes in TCGA cohort by Cytoscape software (online supplementary table 2). Unexpectedly, ClueGO analysis showed that GABRP was extensively involved multiple inflammatory responses, including positive regulation of immune system process, activation of innate immune response, regulation of cytokine production, regulation of leucocyte migration, positive regulation of response to stimulus and positive regulation of nuclear factor κ B (NF- κ B) transcription factor activity, indicating a regulatory role of GABRP in PDAC immune microenvironment (figure 3A). Next, to characterise the potential immune components in the tumour microenvironment affected by GABRP, an immunome compendium was built by publicly available data from purified immune cell subsets.^{20,21} We investigated both innate immune cells (mast cells, macrophages and natural killer cells) and adaptive immune cells [B, T helper 1 (Th1) and CD8+ T] as well as the cytotoxic cells (figure 3B). Intriguingly, we observed that GABRP was specifically associated with the gene signatures of macrophages. In contrast, no obvious correlation was found in the expression of GABRP and other immune component-related genes (figure 3B). To test whether GABRP contributes to macrophage recruitment, the chemotactic assay was performed in phorbol myristate acetate (PMA)-primed THP-1 cells and primary monocytes. After treatment for 24 hours, conditional medium (CM) from sh-GABRP BxPC-3 and PANC-1 cells had reduced capacity to recruit macrophages compared with CM from sh-Ctrl cells (online supplementary figure 2A, B). In addition, CM from high GABRP-expressing cells showed enhanced chemotactic ability than that from low GABRP-expressing cells, suggesting that GABRP might contribute to macrophage infiltration in PDAC (online supplementary figure 2C, D). Immunohistochemical analysis of an independent sample cohort also revealed a positive association between GABRP expression and number of CD68+ and CD163+ macrophages in PDAC tissues (online supplementary figure 3A, B). In addition, GABRP knockdown resulted in reduced macrophage infiltration in the *in vivo* animal models (online supplementary figure

3C). Thus, GABRP might exhibit a regulatory role on macrophage infiltration in PDAC.

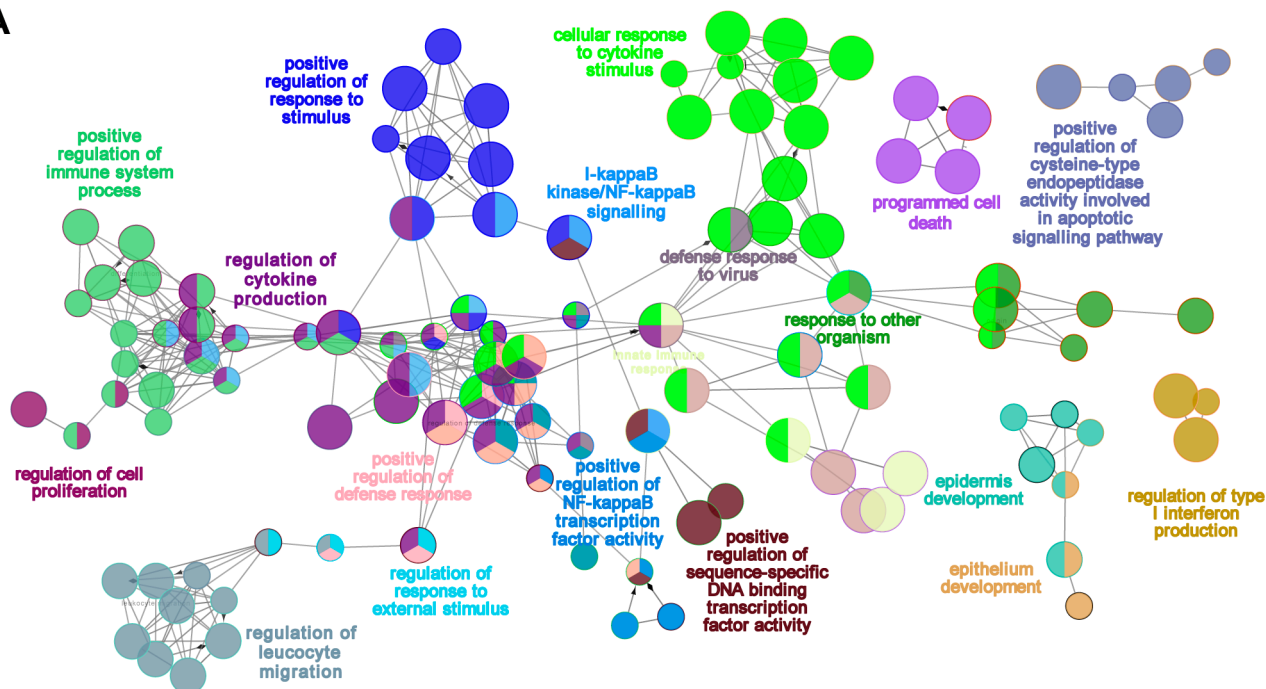
GABRP promotes tumour growth in a macrophage-dependent manner

To demonstrate whether the oncogenic roles of GABRP in PDAC are macrophage-dependent, we investigated the susceptibility of macrophages to liposomal clodronate treatment. To address this issue, we overexpressed *Gabrp* in a mouse PDAC cell line Panc-02, which showed faint GABRP protein expression. The overexpression efficiency is shown in figure 4A. Next, we performed orthotopic xenograft study with *Gabrp*-overexpressing and control cell lines injected in the pancreas of C57BL/6J mice. The mice were pretreated with clodronate liposomes or phosphate-buffered saline (PBS) liposomes for 2 weeks and received treatment for three additional weeks after tumour cell injection (figure 4B). Treatment with clodronate liposomes was sufficient to decrease the number of F4/80+ macrophages in pancreas tumours, liver and lung tissues of C57BL/6J mice (figure 4C). Meanwhile, we also noticed more macrophage infiltration in *Gabrp*-overexpressing group compared with the control group (figure 4C). By using *in vivo* imaging system, we revealed that increased tumour burden induced by *Gabrp*-overexpressing was largely attenuated in mice that had been treated with liposomal clodronate (figure 4D), suggesting that the tumour-promoting effects of GABRP in PDAC, at least in part, are mediated by macrophage infiltration in the tumour microenvironment.

CXCL5 and CCL20 mediate the promoting role of GABRP in macrophage infiltration

To identify the chemokines regulated by GABRP that promote macrophage infiltration, the chemokine profiles of CM from sh-Ctrl and sh-GABRP BxPC-3 cells were analysed using the RayBio Human Cytokine Antibody Array. Two cytokines, CXCL5 and CCL20, were significantly decreased in the CM of sh-GABRP cells compared with sh-Ctrl cells (figure 5A). Real-time PCR showed that GABRP knockdown drastically inhibited the mRNA level of CXCL5 and CCL20 in both BxPC-3 and PANC-1 cells (figure 5B). ELISA assay further confirmed the reduction of CXCL5 and CCL20 in the CM of sh-GABRP cells (figure 5C). In addition, opposite results were observed by overexpression strategies (online supplementary figure 4). Next, we evaluated whether these two chemokines are responsible for macrophage infiltration *in vitro*. As shown in figure 5D, stimulation of PMA-primed THP-1 cells with recombinant CXCL5 or CCL20 showed enhanced migratory ability and this promotive effect was largely compromised by addition of neutralising anti-CXCL5 and anti-CCL20 antibody, indicating that both CXCL5 and CCL20 contribute to macrophage recruitment. Consistently, neutralisation of CXCL5 and CCL20 in the CM from BxPC-3 and PANC-1 cells also suppressed the migratory potential of THP-1 cells (figure 5E). Given that THP-1 cells are transformed and do not perform the physiological functions like normal monocytes, we therefore studied the migratory ability of human peripheral monocytes following treatment with each stimulation and related blocking conditions. Similarly, these results were also reproduced in human monocytes (figure 5D, E). Since CXCL5 and CCL20 bind to the surface of monocytes/macrophages and activate them, we next measured their cell surface receptor CXCR2 and CCR6 by immunofluorescence. Indeed, CXCR2 and CCR6 presented in the THP-1 cells and human monocytes, and intensely expressed in the macrophages infiltrated in the PDAC tissues (online supplementary figure

A



B

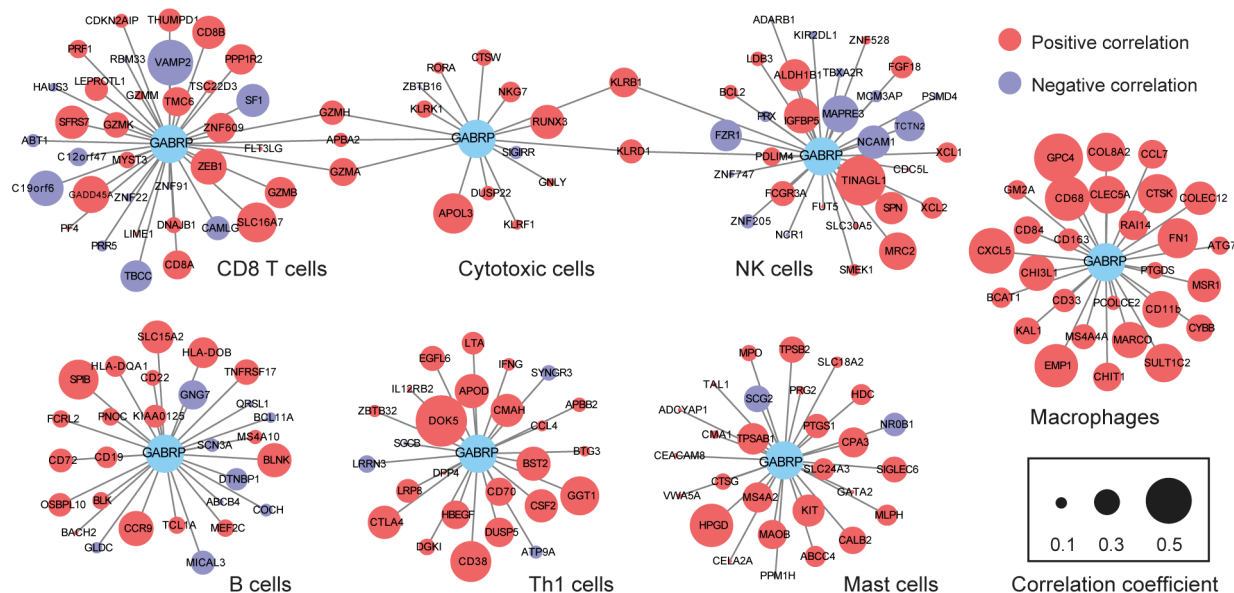


Figure 3 Gamma-aminobutyric acid type A receptor pi subunit (GABRP) correlates with macrophage infiltration in pancreatic ductal adenocarcinoma (PDAC). (A) ClueGO analysis of the top 300 genes having the highest correlation with *GABRP* by CytoScape software. Enriched pathways are shown as nodes interconnected based on the κ score. (B) The correlation between *GABRP* and specific gene signatures of different immune cells. Node size represents the association p value between the neighbour gene and *GABRP*. NF, nuclear factor; NK, natural killer; Th, T helper.

5A). Genetic inhibition of CXCR2 or CCR6 remarkably inhibited the migratory ability of THP-1 cells and human monocytes (online supplementary figure 5B–D). Together, we conclude that alterations in CXCL5 and CCL20 levels induced by GABRP are indeed responsible for macrophage infiltration in PDAC.

GABRP modulates intracellular Ca^{2+} through interaction with KCNN4

GABA_A receptor signalling leads to chloride (Cl^-) influx and subsequent calcium mobilisation. Actually, there was an abundant source for GABA derived from tumour cells and nerve fibres in

PDAC tissues (online supplementary figure 6A). However, GABA treatment did not induce calcium mobilisation in PDAC cells or affected cell proliferation and invasion as revealed by FLUO-3 AM staining, cell viability assay and transwell assay (online supplementary figure 3B–D). Notably, GABRP-silenced BxPC-3 and PANC-1 cells had lower $[\text{Ca}^{2+}]_i$ compared with control cells (figure 6A, B). In addition, chelation of intracellular Ca^{2+} by BAPTA suppressed the mRNA levels of CXCL5 and CCL20 in BxPC-3 and PANC-1 cells (online supplementary figure 3E), indicating that altered calcium signalling might be responsible for changes in chemokine expression in PDAC. Based on the

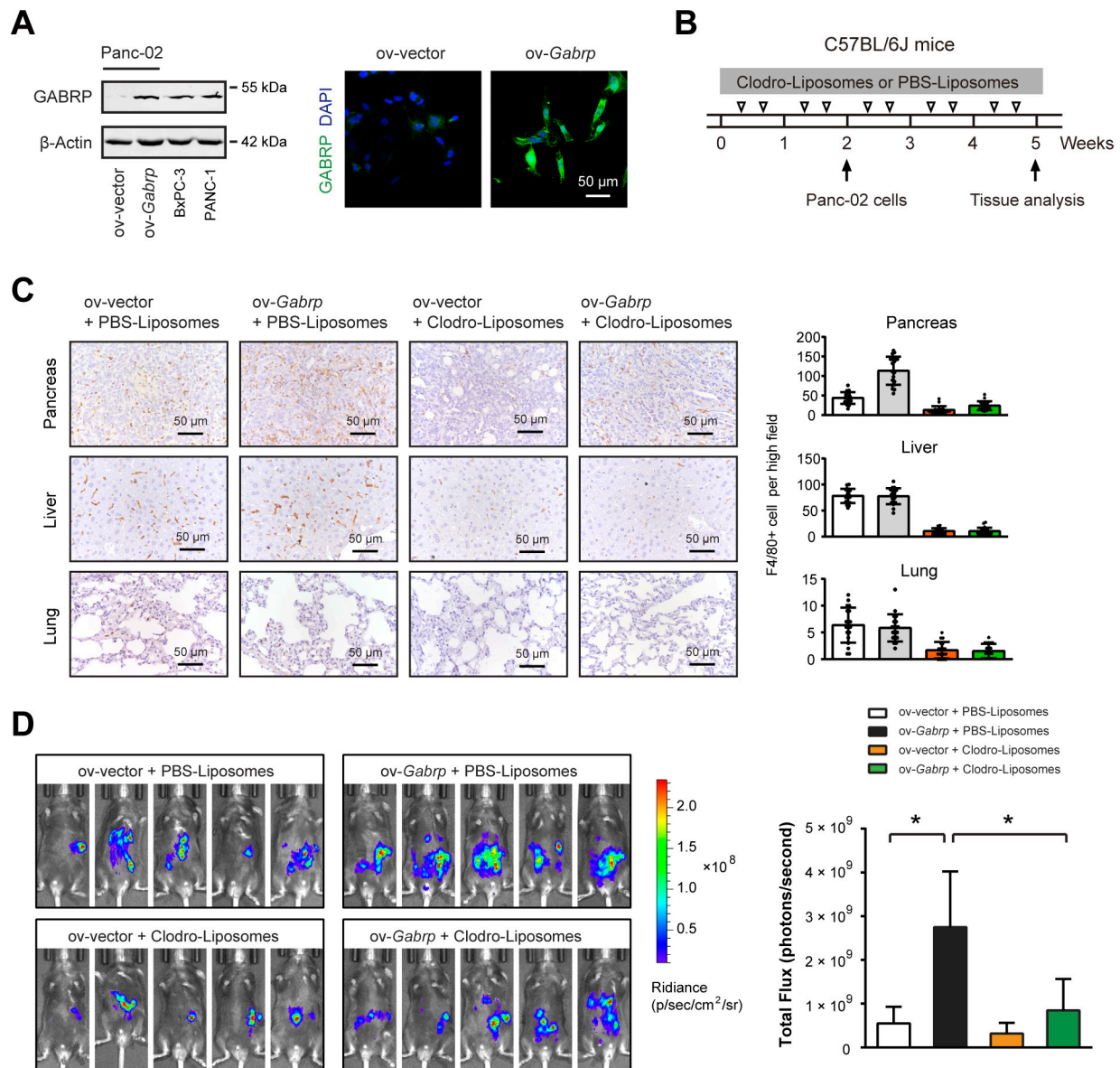


Figure 4 Gamma-aminobutyric acid type A receptor pi subunit (GABRP) promotes tumour growth in a macrophage-dependent manner. (A) Western blot analysis and immunofluorescence analysis of *Gabrp* overexpression efficiency in Panc-02 cells. (B) Treatment scheme depicting intraperitoneal application of clodronate liposomes or phosphate-buffered saline (PBS) liposomes in C57BL/6J mice for 5 weeks. Panc-02 cells were orthotopically administered at the end of week 2 and animals were sacrificed 3 weeks later. (C) Representative immunohistochemistry of F4/80+ cells in sections from pancreas tumours, liver and lung tissues obtained from C57BL/6J mice treated with clodronate liposomes or PBS liposomes. Scale bar: 50 μ m. (D) In vivo imaging analysis of ov-vector and ov-Gabrp tumours from clodronate-liposome or PBS-liposome-treated mice (n=5 per group). *P<0.05.

evidence listed below: (1) GABRP-mediated calcium mobilisation in PDAC was GABA-independent; (2) Ca^{2+} entries can be modulated by potassium (K^+) channels^{22–24}; (3) K^+ channels must accompany Cl^- channel to maintain electroneutrality and they commonly co-localise on the cell membrane in lipid rafts; (4) GABRP controls the sensitivity of the Cl^- channel GABA_A , we speculated that GABRP regulates intracellular Ca^{2+} through interaction with K^+ channels. To test this hypothesis, we mined the TCGA cohort and found that Ca^{2+} -activated K^+ channel KCNN4 was the most abundantly expressed K^+ channel in PDAC and had the highest correlation coefficient with GABRP (online supplementary figure 7A, B). Overexpressed KCNN4 predicts a poor prognosis in patients with PDAC and modulates pancreatic cancer cell migration, invasion and proliferation.^{25, 26} To investigate whether the KCNN4 channels are functional in

PDAC cells, we tested the intracellular Ca^{2+} activation and pharmacological properties of the channel by electrophysiological patch-clamp measurement. As shown in online supplementary figure 7C, the whole-cell K^+ current was almost undetectable when the intracellular free Ca^{2+} was around 0. By contrast, the fast perfusion with 1 μM Ca^{2+} internal solution significantly increased the ionic current which can be blocked by extracellular application of KCNN4 blocker TRAM34, suggesting that KCNN4 is functional in BXP-3 cells. Similarly, the KCNN4 current was also confirmed in PANC-1 cells (online supplementary figure 7D). Next, by co-immunoprecipitation assay and proximity ligation assay, we confirmed the physical interaction between KCNN4 and GABRP (figure 6B, C). Immunofluorescence analysis showed that KCNN4 and GABRP were localised at the plasma membrane (figure 6D). Intriguingly, K^+ conductance

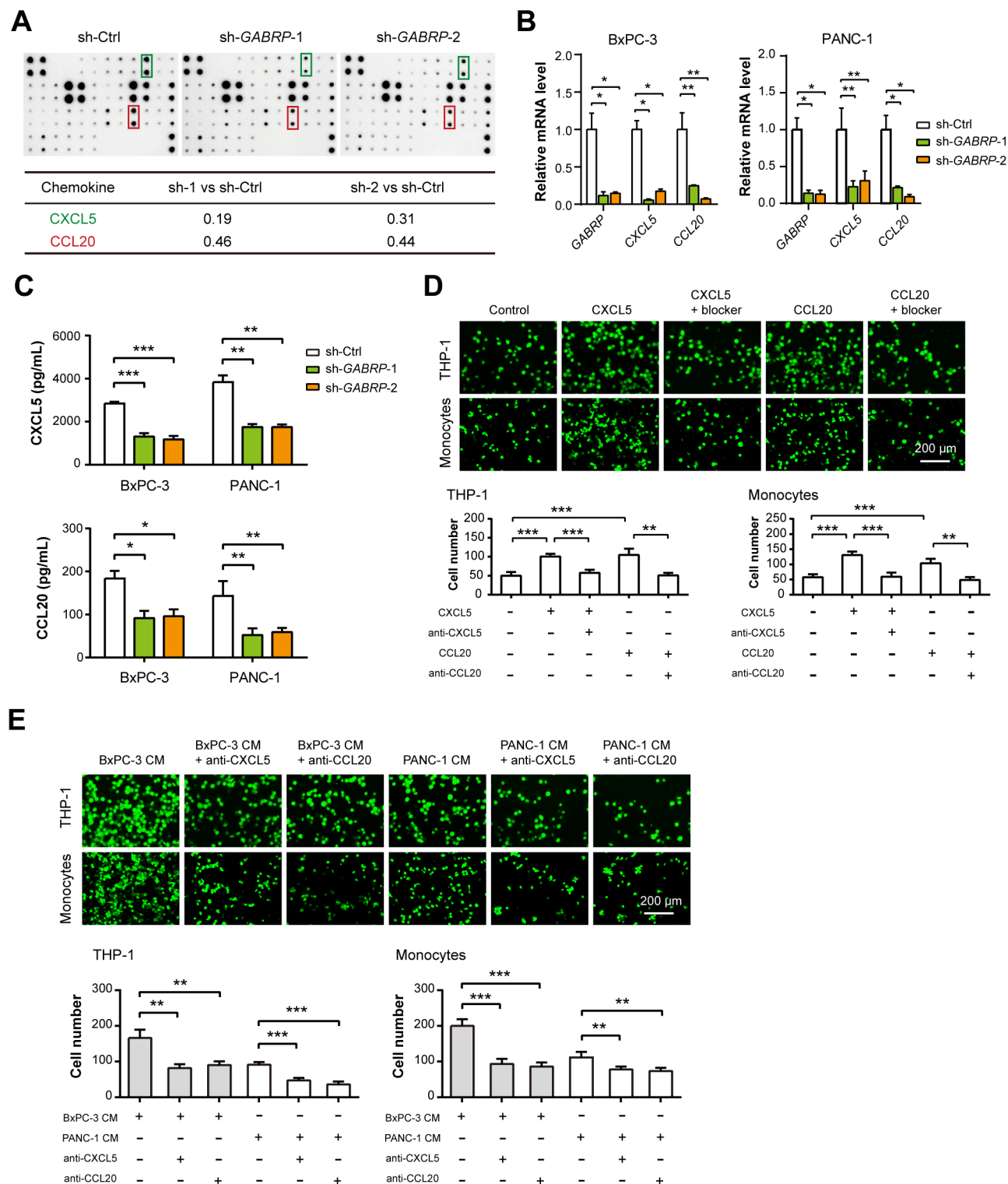


Figure 5 CXCL5 and CCL20 mediate the promoting role of macrophage infiltration induced by gamma-aminobutyric acid type A receptor pi subunit (GABRP). (A) Chemokine array of the conditional medium (CM) from short hairpin (sh)-Ctrl and sh-GABRP BxPC3 cells. A table summarising the relative signal intensity of indicated chemokines is presented in the lower left corner. (B) Real-time qPCR analysis of the effect of GABRP knockdown on the CXCL5 and CCL20 mRNA level in BxPC-3 and PANC-1 cells (n=3). (C) ELISA analysis of the CXCL5 and CCL20 level in the CM from sh-Ctrl and sh-GABRP BxPC-3 and PANC-1 cells (n=3). (D) Effects of rCXCL5 and rCCL20 on the migratory ability of phorbol myristate acetate (PMA)-primed THP-1 cells and human monocytes (n=6). (E) Pancreatic ductal adenocarcinoma (PDAC) cell culture supernatants were neutralised with anti-CXCL5 antibody and anti-CCL20 antibody and the migratory ability of PMA-treated THP-1 cells and human monocytes was analysed (n=6). *P<0.05, **p<0.01, ***p<0.001.

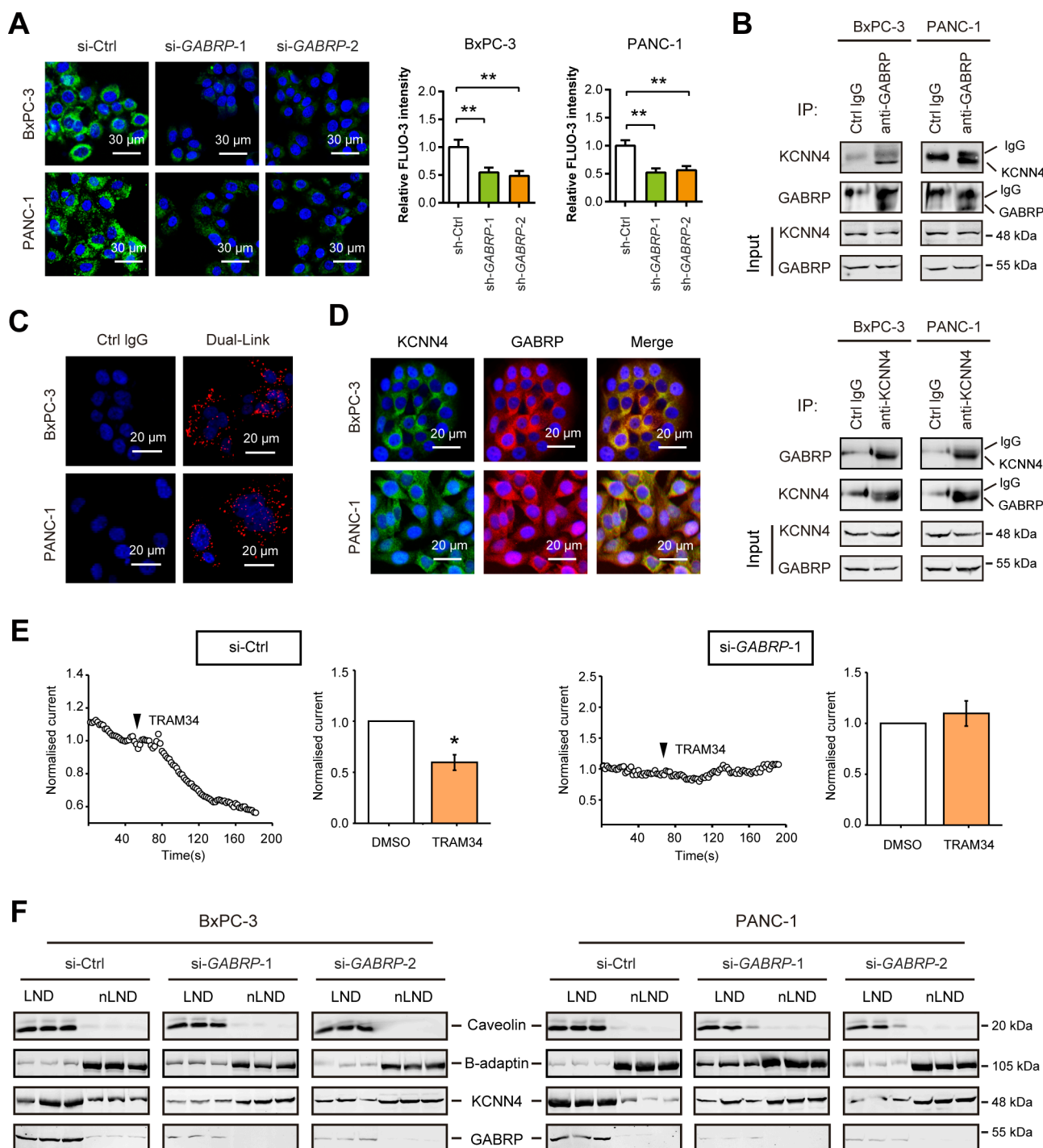


Figure 6 Gamma-aminobutyric acid type A receptor pi subunit (GABRP) modulates intracellular Ca^{2+} through interaction with KCNN4. (A) The effects of *GABRP* knockdown on the free cytosolic calcium ($[\text{Ca}^{2+}]$) concentration was measured using FLUO-3 AM staining. (B) Co-immunoprecipitation analysis of the interaction between GABRP and KCNN4 expression in the BxPC-3 and PANC-1 cells. (C) In situ proximity ligation assay analysis of the interaction between GABRP and KCNN4. (D) Co-immunofluorescence analysis of GABRP and KCNN4 expression in the BxPC-3 and PANC-1 cells. (E) Whole-cell patch clamp recordings of short hairpin (sh)-Ctrl and sh-*GABRP-1* pancreatic ductal adenocarcinoma (PDAC) cells with $1 \mu\text{M}$ free Ca^{2+} in pipette solution in the absence or presence of KCNN4 inhibitor TRAM-34. (F) Distribution of KCNN4 and GABRP in membrane fraction isolated from BxPC-3 and PANC-1 cells treated with sh-Ctrl or sh-GABRP. Caveolin-1 was used as a marker of lipid nanodomains and β -adapitin was used as non-lipid nanodomains marker. LND, lipid nanodomains; nLND, non-lipid nanodomains. * $P < 0.05$, ** $P < 0.01$.

after GABRP knockdown was insensitive to TRAM34, indicating the regulatory role of GABRP on KCNN4 channel (figure 6E). The activation of KCNN4 channel results in cell membrane hyperpolarisation and induces calcium entry by increasing the electrochemical driving force for calcium.^{23 24} Expectedly,

siRNA-mediated KCNN4 knockdown led to pronounced decrease in intracellular Ca^{2+} level and *CXCL5* and *CCL20* levels in BxPC-3 and PANC-1 cells (online supplementary figure 8A-C). To further confirm the regulatory role of KCNN4/ Ca^{2+} axis on *CXCL5* and *CCL20* expression, we analysed the effects

of KCNN4 activator 1-EBIO and inhibitor TRAM34 in BxPC-3 and PANC-1 cells. Activation of the KCNN4 channel by 100 μ M 1-EBIO led to a slight increase in CXCL5 and CCL20 levels indicative of these channels are already in the active state (online supplementary figure 8D). Moreover, inhibition of the KCNN4 channel by treatment with 20 μ M TRAM-34 significantly reduced CXCL5 and CCL20 levels in BxPC-3 and PANC-1 cells (online supplementary figure 8E). Given various ion channels have been shown to localise to lipid raft, we hypothesised that GABRP may regulate KCNN4 membrane localisation. To address this, membrane fractionation experiment was performed. The result showed that GABRP-silencing totally displaced KCNN4 outside of lipid rafts (figure 6F). Disruption of lipid raft by M β CD (a lipid raft destructor) was sufficient to reduce cholesterol content and constitutive Ca²⁺ entry (online supplementary figure 9A, B). M β CD treatment significantly blocked CXCL5 and CCL20 levels (online supplementary figure 9C). In addition, knockdown of GABRP, knockdown of KCNN4 and treatment with M β CD could all inhibit PDAC cell viability, promote cell apoptosis and suppress invasive potential (online supplementary figure 10). Collectively, these results above suggest that GABRP may act as a chaperone protein and regulate the activity of KCNN4 channel to induce Ca²⁺ signalling in PDAC cells.

NF- κ B pathway is responsible for Ca²⁺-dependent CXCL5 and CCL20 expression

To elucidate the downstream signalling pathways responsible for the Ca²⁺-mediated CXCL5 and CCL20 expression, the three well-known downstream transcriptional factors of Ca²⁺, cAMP responsive element-binding protein (CREB), nuclear factor of activated T cells 1 (NFAT1) and NF- κ B,²⁷ were studied by luciferase reporter assay. We found that GABRP knockdown significantly decreased NF- κ B but not CREB or NFAT1 transcriptional activity in both BxPC-3 and PANC-1 cells (figure 7A). Similarly, genetic inhibition of KCNN4 induced a specific decrease of NF- κ B-dependent luciferase activity (figure 7B). NF- κ B is the widely reported transcription factor for CXCL5 and CCL20. Consistently, GABRP-associated genes were profoundly implicated in NF- κ B signalling pathway (figure 2A). Furthermore, the phosphorylation level of p65 Ser536, a critical I- κ B kinase complex (IKK) complex-mediated modification for NF- κ B activity, in PDAC cells regulated by GABRP-KCNN4 axis was investigated. The result revealed that the phosphorylation level of NF- κ B inhibitor α (I κ B α) and p65 was markedly reduced by knockdown of GABRP or KCNN4 (figure 7C), which can be reversed by ectopic expression of constitutively active IKK β (IKK β CA). Next, to evaluate the dependence of NF- κ B in chemokine production, BxPC-3 and PANC-1 cells were treated with the small-molecule NF- κ B inhibitor Bay11-7082 (I κ B α phosphorylation inhibitor). As expected, Bay11-7082 led to NF- κ B suppression and significantly mitigated the mRNA and protein level of CXCL5 and CCL20 in both two cell lines (figure 7D). Additionally, ectopic expression of IKK β CA largely rescued the decreased NF- κ B-responsive reporter activity and CXCL5 and CCL20 level induced by GABRP and KCNN4 inhibition (figure 7E). To decipher the GABRP-NF- κ B-CXCL5/CCL20 axis in vivo, we performed immunohistochemistry analysis in the tumours tissues from subcutaneous and orthotopic xenograft model. As a result, we found that GABRP knockdown significantly reduced CXCL5 and CCL20 protein expression in subcutaneous xenograft model, while overexpression of *Gabrp* remarkably promoted CXCL5 and CCL20 protein levels in orthotopic xenograft tumour tissues (online supplementary

figure 11A). In addition, *GABRP* and *KCNN4* expression were closely associated with the *CXCL5* and *CCL20* levels in TCGA cohort (online supplementary figure 11B, C). Collectively, these results suggest that Ca²⁺-mediated NF- κ B activation is necessary for the induction of inflammatory chemokine expression induced by GABRP (figure 7F).

DISCUSSION

Neurotransmitters and neurotrophic factors are capable of facilitating PDAC cell tumourigenicity via neurotransmitter receptors and neuroactive ligand receptors.⁹ PDAC cells can secrete immunomodulatory agents to induce an immunosuppressive microenvironment, which is thought to promote tumour progression and invasiveness.^{28 29} In this study, we identified a novel neurotransmitter-independent immunomodulatory role of GABRP in PDAC. Dysregulated GABRP predicts a poor prognosis, and contributes to tumour progression and macrophage infiltration in PDAC.

Previously, several neurotransmitter receptors have been demonstrated to be differentially expressed in PDAC cells and contribute to the malignant phenotypes of PDAC.^{30–32} For example, expression of dopamine receptor D2 (DRD2) is increased in PDAC and inhibition of DRD2 slows tumour growth.³¹ Recently, we have reported that serotonin receptor HTR2B is overexpressed in PDAC cells and activation of HTR2B signalling enhances Warburg effect to facilitate cell survival under metabolic stress.³⁰ In the current study, we revealed that many neurotransmitter receptors and neuroactive ligand receptors are differentially expressed in PDAC, especially GABRP, a subunit of GABA_A receptor with the largest fold change. Unlike other GABA_A subunits, GABRP is not enriched in the brain tissue, but is detectable in multiple non-neuronal tissues, such as the uterus and mammary gland.^{13 33} Consistently, specific expression of GABRP was also detected in the islets of normal pancreas. GABA receptors have been detected in many tumour tissues and exert regulative effects in cancer cell proliferation and migration.^{34 35} Our findings that GABRP associates with tumour growth and metastasis are supported by previous observations at basal-like breast cancer³⁶ and oral squamous cell carcinoma.³⁷ Despite the abundant source of GABA derived from nerve fibres and cancer cells in the tumour microenvironment, our results suggest a GABA-independent oncogenic role of GABRP in PDAC. Different from our findings, Takehara *et al* reported that GABA stimulates pancreatic cancer growth through the functional chloride channel which might be formed by GABRP.¹⁵ However, the levels of other main subunits of GABA_A receptor such as α , β and γ subunits were undetectable or faintly expressed in PDAC tissues and GABRP cDNA did not express binding sites for the GABA_A receptor ligands,¹³ indicating that GABRP-overexpressing PDAC cells without other GABA_A receptor subunits fail to assemble homomeric chloride channels that are gated by their ligands. Therefore, we conclude that GABRP-mediated oncogenic functions might be mediated by other mechanisms rather than forming functional chloride channel.

Next, we uncovered that GABRP is closely associated with macrophage infiltration in PDAC. The infiltration of immune cells is, among other events, of great importance for antitumour immune response, and is known to have a strong impact on the outcome of human cancers.^{38 39} Macrophages are a prominent component of the leucocyte infiltrate in human cancers and play a major role in orchestrating the cancer-related inflammation.⁴⁰ Tumour-derived factors stimulated the recruitment and differentiation of macrophages.⁴¹ In this study, GABRP expression correlates with the majority of gene signatures related

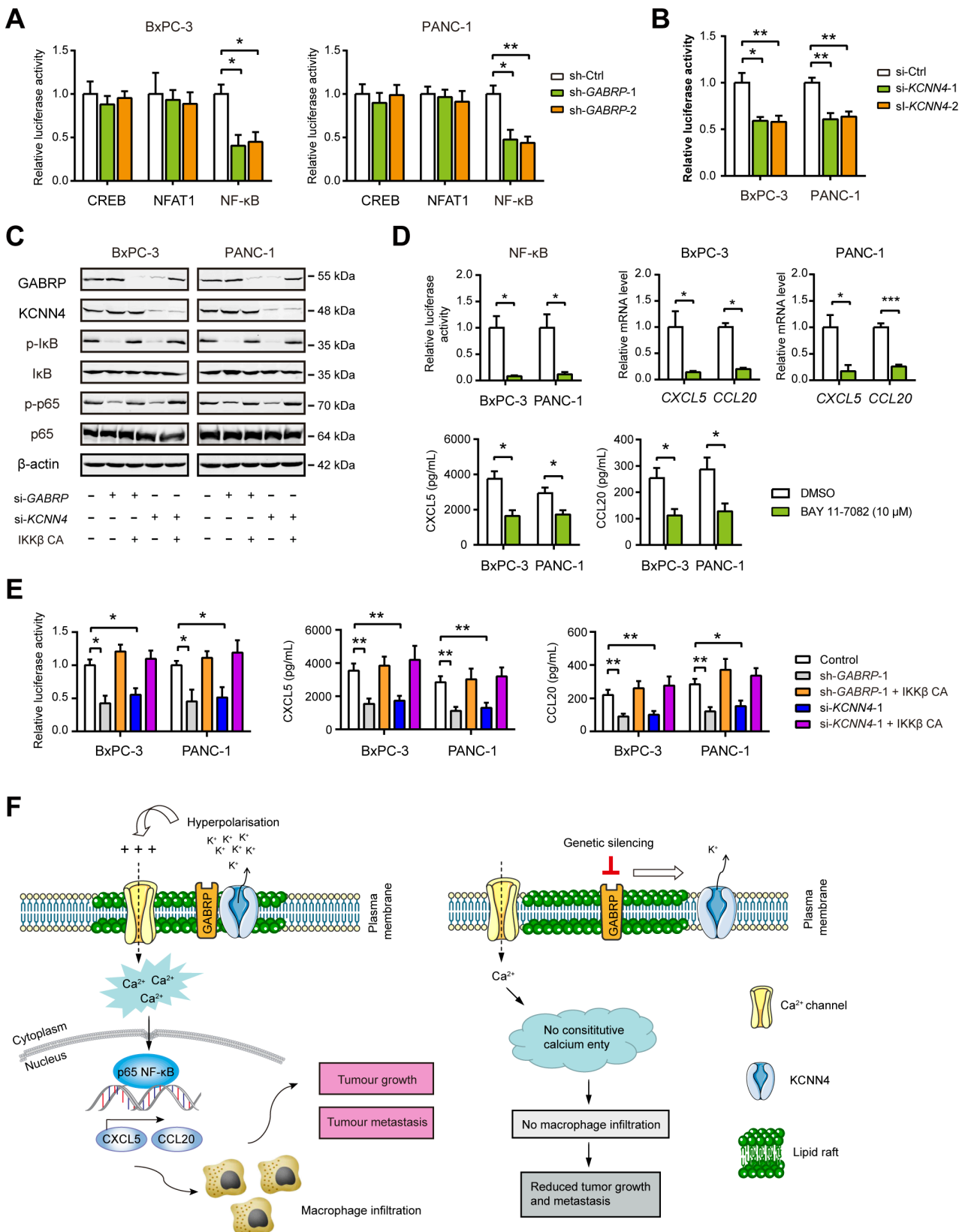


Figure 7 Nuclear factor κ B (NF- κ B) pathway is responsible for Ca^{2+} -dependent CXCL5 and CCL20 expression. (A) Effects of gamma-aminobutyric acid type A receptor pi subunit (GABRP) knockdown on the transcriptional activity of cAMP responsive element-binding protein (CREB), nuclear factor of activated T cells 1 (NFAT1) and NF- κ B in BxPC-3 and PANC-1 cells ($n=3$). (B) Effects of KCNN4 knockdown on the transcriptional activity of NF- κ B in BxPC-3 and PANC-1 cells ($n=3$). (C) Western blot analysis of the effect of GABRP and KCNN4 knockdown on NF- κ B pathway activity. (D) Effect of NF- κ B inhibitor Bay11-7082 (10 μM) on CXCL5 and CCL20 expression in BxPC-3 and PANC-1 cells ($n=3$). (E) Effect of IKK β on the NF- κ B transcriptional activity, CXCL5 and CCL20 expression in GABRP or KCNN4-silenced PDAC cells ($n=3$). (F) A schematic diagram shows the mechanism by which GABRP regulates macrophage infiltration in pancreatic cancer and promotes tumour progression. * $P<0.05$, ** $p<0.01$, *** $p<0.001$.

to macrophages and are closely associated with macrophage density, suggesting the tumour-promoting effects of GABRP in PDAC might be linked to the macrophage infiltration. There is a strong association between increased macrophage density and poor prognosis.¹⁶ Macrophages can suppress Th1 cell and cytotoxic T lymphocyte antitumour responses, contribute to matrix remodelling and facilitating tumour cell migration and invasion and promote tumour angiogenesis and growth.^{42–44} In line with this, we showed that deletion of macrophages by liposomal clodronate largely compromises the growth-promoting effects of GABRP in orthotopic xenograft model, suggesting that the oncogenic roles of GABRP in PDAC are largely dependent on macrophages.

Subsequently, we for the first time identified a novel mechanism that GABRP interacts with KCNN4 and promotes macrophage infiltration by activating the $\text{Ca}^{2+}/\text{NF-}\kappa\text{B}/\text{CXCL5-CCL20}$ axis. KCNN4, also known as KCa3.1 , is found to be highly expressed in many cancer types and implicated in the promotion of cell invasion and cell proliferation. KCNN4-induced calcium signalling process is critically involved in the malignant phenotypes of various cancers such as melanoma, prostate cancer, pancreatic cancer and glioblastoma.^{24 25 45 46} Genetic or pharmacological inhibition of KCNN4 remarkably suppresses prostate cancer cell proliferation and migration and melanoma cell migration.^{24 46} Importantly, both processes are mediated by an increase in the intracellular Ca^{2+} concentration. In this study, we identified GABRP as an interactor of KCNN4 in lipid rafts, of which deletion reduces $[\text{Ca}^{2+}]_i$. GABRP may act as a channel modulator of KCNN4, just like its function in the GABA_A receptor. However, the detailed physical association of KCNN4 channel with GABRP warrants further investigations. Intracellular Ca^{2+} plays a crucial role in almost all cellular processes and the diversity of Ca^{2+} signalling are relevant in multiple aspects of tumour progression.²⁷ The tightly regulated $[\text{Ca}^{2+}]_i$ in the form of waves, spikes or oscillations selectively activate the transcription factors for gene expression, cell migration and cell proliferation.^{47 48} As the well-characterised downstream targets of Ca^{2+} , calmodulin kinases and calcineurin turn on the transcription factors such as NFAT and NF- κB . In the nucleus, Ca^{2+} can be delivered by Ca^{2+} /calmodulin-dependent protein kinase II subunit γ (CaMKII γ) to stimulate the phosphorylation of CREB and gene expression.^{27 49} Consistently, we revealed that GABRP-KCNN4 complex leads to a specific Ca^{2+} -dependent activation of NF- κB , which further facilitates CXCL5 and CCL20 transcription to induce macrophage infiltration in PDAC. Macrophage chemotaxis is predominantly regulated by several growth factors and chemokines in PDAC such as CCL-2 and CSF-1.^{50 51} However, CXCL5 and CCL20 are also profoundly implicated in the recruitment and function of macrophage and subsequent tumour progression.^{52 53} For example, CXCL5-deficient prostate cancer exhibits reduced macrophage infiltration and tumour growth.⁵² Thus, our data, as a proof of principle, provides several potential molecular and immune targets for cancer therapeutics in PDAC.

Last but not least, our investigation focused on GABRP, to some extent, is not the best model reflecting the interactions between nerve fibres and cancer cells in PDAC. Occasionally, we revealed an unexpected neurotransmitter-independent immunomodulatory function of GABRP in PDAC. However, the notion that nerves drive the tumour initiation and progression are well deciphered by two excellent reports.^{54 55} In prostate cancer, sympathetic nerves promote early stages of tumourigenesis, whereas parasympathetic nerve fibres promote cancer dissemination.⁵⁴ In gastric cancer, surgical or pharmacological denervation of the stomach or genetic knockout of the muscarinic acetylcholine

M3 receptor reduces tumour incidence and progression.⁵⁵ Of note, many immune cells express neurotransmitter receptors,⁵⁶ and GABA/ GABA_A receptor-mediated immunomodulatory functions have been unveiled in several immune cells, such as T cell, dendritic cell and microglia.⁵⁷ Thus, whether GABA modulates the inflammatory response in the tumour microenvironment directly by targeting the immune cells in PDAC warrants further study.

In conclusion, our study showed that dysregulated neurotransmitter receptor GABRP promotes cancer aggressiveness and correlates macrophage infiltration in PDAC and provides a new insight into the roles of neurotransmitter receptor in immune modulation. Meanwhile, we uncovered an unprecedented mechanism that GABRP interacts with KCNN4 to modulate intracellular calcium signalling. These findings above demonstrate that point towards the GABRP/KCNN4/ $\text{Ca}^{2+}/\text{NF-}\kappa\text{B}/\text{CXCL5-CCL20}/\text{macrophage}$ axis as potential therapeutic target in PDAC.

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Correction notice This article has been corrected since it published Online First. Figure 4 has been replaced with the correct figure and the equal contribution statement has been added.

Acknowledgements The authors would like to thank Professor Jing Xue for providing the Panc-02 cells. The authors would also like to thank Dr Jun-Ping Ao for technical help.

Contributors S-HJ, L-LZ and R-KL carried out in vitro cell experiments, in vivo experiments, manuscript preparation and statistical analysis; QY contributed to bioinformatics analysis; J-YY, L-YT, Y-SJ and RH provided clinical specimens and made clinical pathology evaluations; MZ, CZ and J-YY performed electrophysiological testing; L-PH, JL and Y-HW contributed to animal breeding; F-YD, MD, QL, G-AT, X-XZ, X-YC and SH performed western blot analysis and IHC analysis; X-MY, Y-LZ, H-ZN, LZ, X-LZ and K-QL provided critical review; J-RG, S-WH, Y-W and Z-GZ conceived, designed, supervised, analysed and interpreted the study and provided critical review.

Funding This work was supported by grants from National Natural Science Foundation of China (81871923, 81802890, 81872242, 81502382, 81500461 and 81761138045), the Natural Science Foundation of Shanghai (17ZR1428300 and 18ZR1436900), Shanghai Municipal Commission of Health and Family Planning (201740105), Shanghai Municipal Education Commission—Gaofeng Clinical Medicine Grant Support (20181708), Shanghai Municipal Health Bureau (2018BR32) and the State Key Laboratory of Oncogenes and Related Genes (no 91-17-14).

Competing interests None declared.

Patient consent for publication Obtained.

Ethics approval The study was approved by the Research Ethics Committee of Ren Ji Hospital, School of Medicine, Shanghai Jiao Tong University. The experiments were approved by the Research Ethics Committee of East China Normal University.

Provenance and peer review Not commissioned; externally peer reviewed.

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